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Two-way ANOVA — post hoc analysis, what is the difference between TukeyHSD and T-test

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Wei hung Pan

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ANOVA (analysis of the variance) is a statistical method to test the equality of the group means. Furthermore, 2-way anova is used if there are 2 factors.

In this story, we will use in-vivo gene expression data with biological replicates as the example for the 2-way ANOVA analysis.

Data Information

We have a series of gene expression value on 3 different concentration RNA (Low, Medium and High) and 3 mice.

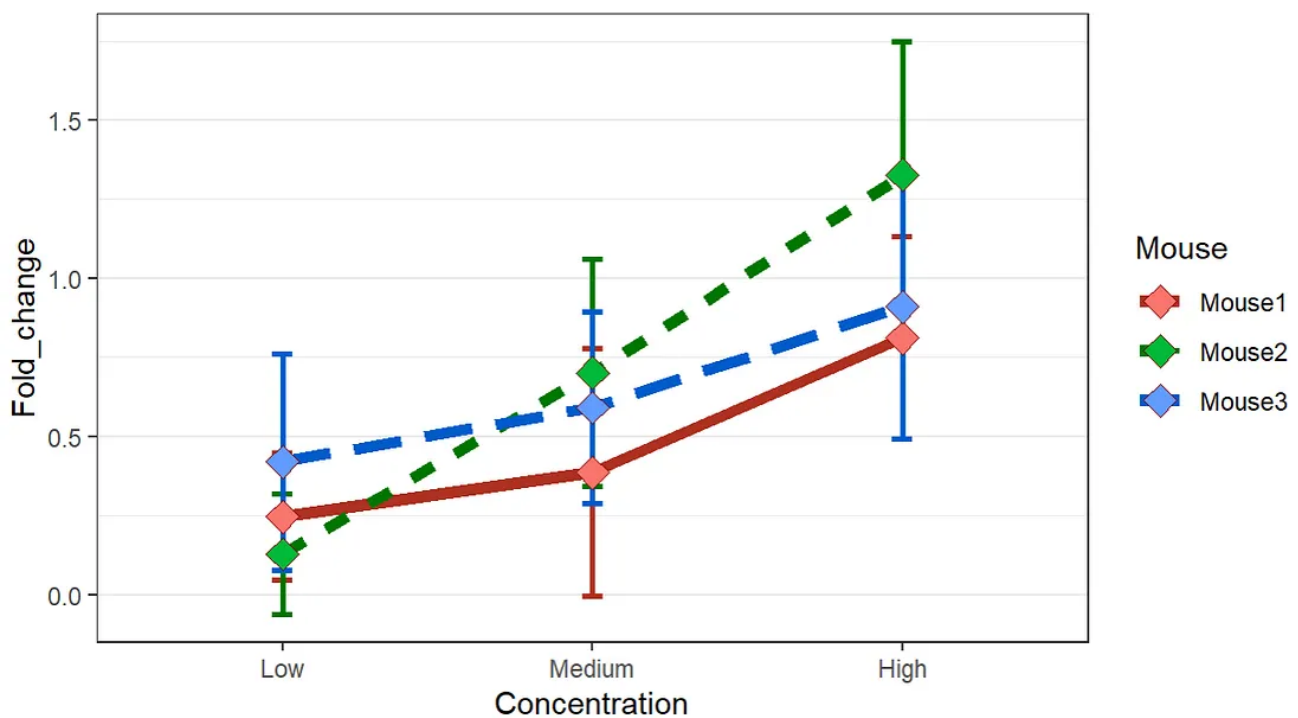
Fold_change	Concentration	Mouse
-0.2638956706	High	Mouse1
1.9741910299	Medium	Mouse2
0.9199638367	Low	Mouse3
1.2956136068	High	Mouse1
1.0926793416	Low	Mouse2

Raw data

It is a 3 X 3 experiment, means 3 factors by 3 factors.

Visualization

Before we jump into the analysis, visualization is always the first thing we need to do.



Here we can see the fold change increase when the concentration is high.

Fit the model

The same as other statistician, I use R for my analysis

First I tried the full model with interaction

```
two_way_interaction<- aov(Fold_change ~ Mouse * Concentration,
data)
summary(two_way_interaction)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
Mouse	2	2.47	1.233	3.019	0.052	.
Concentration	2	13.43	6.715	16.447	3.8e-07	***
Mouse:Concentration	4	2.38	0.594	1.455	0.219	
Residuals	141	57.57	0.408			

Here we can see from the concentration effect is very obvious, and mouse effect is *nearly* significant (p-value=0.052). The interaction is not significant.

In this scenario, i will re-run the model without the interaction

```
two_way_only_main<- aov(Fold_change ~ Mouse + Concentration, data)
summary(two_way_only_main)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
Mouse	2	2.47	1.233	2.981	0.0538	.
Concentration	2	13.43	6.715	16.243	4.31e-07	***
Residuals	145	59.94	0.413			

We can see the mouse effect is still on edge of significance. I would keep mouse in the model and have more discussion later.

For the model fitting, I will first run the full model, and then adjust the analysis. Because anova also belong to linear model, you can also choose the AIC approach to find the best model to fit the data.

```
library(AICcmodavg)
```

```
model.set <- list(two_way_only_main, two_way_interaction)
model.names <- c("Without interaction", "With interaction")

aictab(model.set, modnames = model.names)
```

Model selection based on AICc:

	K	AICc	Delta_AICc	AICcWt	Cum.Wt	LL
without interaction	6	300.68	0.00	0.81	0.81	-144.05
with interaction	10	303.61	2.93	0.19	1.00	-141.01

The AIC model with the best fit will be listed first. This comparison reveals that the two-way ANOVA without any interactions is the best fit for the data.

Post-hoc analysis

When you use ANOVA to find the main effect is significant, indicate that the main effect is different between groups. However, you might want to know the which group is different than the other group, that is called post-hoc analysis.

The Tukey HSD (“honestly significant difference” or “honest significant difference”) is used often in the post-hoc analysis. At the time, t-tests were being applied to three or more groups, and John Tukey considered this dishonest — hence “honestly significant difference.”

What Is the Tukey HSD Test?

The Tukey HSD test is a way of reporting ANOVA results and determining if the relationship between three independently...

[sciencing.com](https://www.sciencing.com)

```
TukeyHSD(two_way_only_main)
```

Tukey multiple comparisons of means 95% family-wise confidence level

```
Fit: aov(formula = Fold_change ~ Mouse + Concentration, data = data)
```

\$Mouse

	diff	lwr	upr	p adj
Mouse2-Mouse1	0.3128359	0.008325671	0.6173461	0.0425708
Mouse3-Mouse1	0.1799196	-0.124590620	0.4844298	0.3440228
Mouse3-Mouse2	-0.1329163	-0.437426504	0.1715939	0.5569161

\$Concentration

	diff	lwr	upr	p adj
Medium-Low	0.256737	-0.04529608	0.5587700	0.1127257
High-Low	0.717543	0.41550997	1.0195760	0.0000003
High-Medium	0.460806	0.15001661	0.7715955	0.0017147

We can see the difference between Mouse2-Mouse1, High-Low and High-Medium.

What is the different between TukeyHSD and T-test

you might want to ask, what if I compare the 2 groups with simple t-test, what is the difference of Tukey and T-test? the answer:

Multiple correction

Multiple testing refers to **any instance that involves the simultaneous testing of more than one hypothesis**. If decisions about the individual hypotheses are based on the un-adjusted marginal p-values, then there is typically a large probability that some of the true null hypotheses will be rejected.

The Tukey HSD (honest significant difference) takes into account multiple comparisons itself. It's based on the Studentized range distribution of the difference between the maximum and minimum of a set of a number of values

Thus, the calculation is different. In the TukeyHSD, the q-statistics is based on the mean difference divide by the pool standard error of the sum of the means

$$q_s = \frac{Y_A - Y_B}{SE},$$

where $\bar{Y}_A$ is the larger of the two means being compared, $\bar{Y}_B$ is the smaller of the two means being compared, and SE is the standard error of the sum of the means from all groups.

If you perform the t-test, you use the pool standard deviation **only between 2 groups**.

$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p \sqrt{\frac{2}{n}}}$$

Let's try to run the t-test in our data

```
t.test(Fold_change ~ Concentration, data = data %>%
dplyr::filter(!Concentration == "High"))
```

Welch Two Sample t-test

```
data: Fold_change by Concentration
t = -2.7804, df = 87.433, p-value = 0.006647
alternative hypothesis: true difference in means between
group Low and group Medium is not equal to 0
95 percent confidence interval:
-0.56534164 -0.09402149
sample estimates:
mean in group Low mean in group Medium
0.2690074 0.5986890
```

Here we compare Low vs Medium. The t-test showing pvalue is 0.006 , however, if we consider the multiple correction , the adjusted pvalue from TukeyHSD is 0.112 which is not significant.

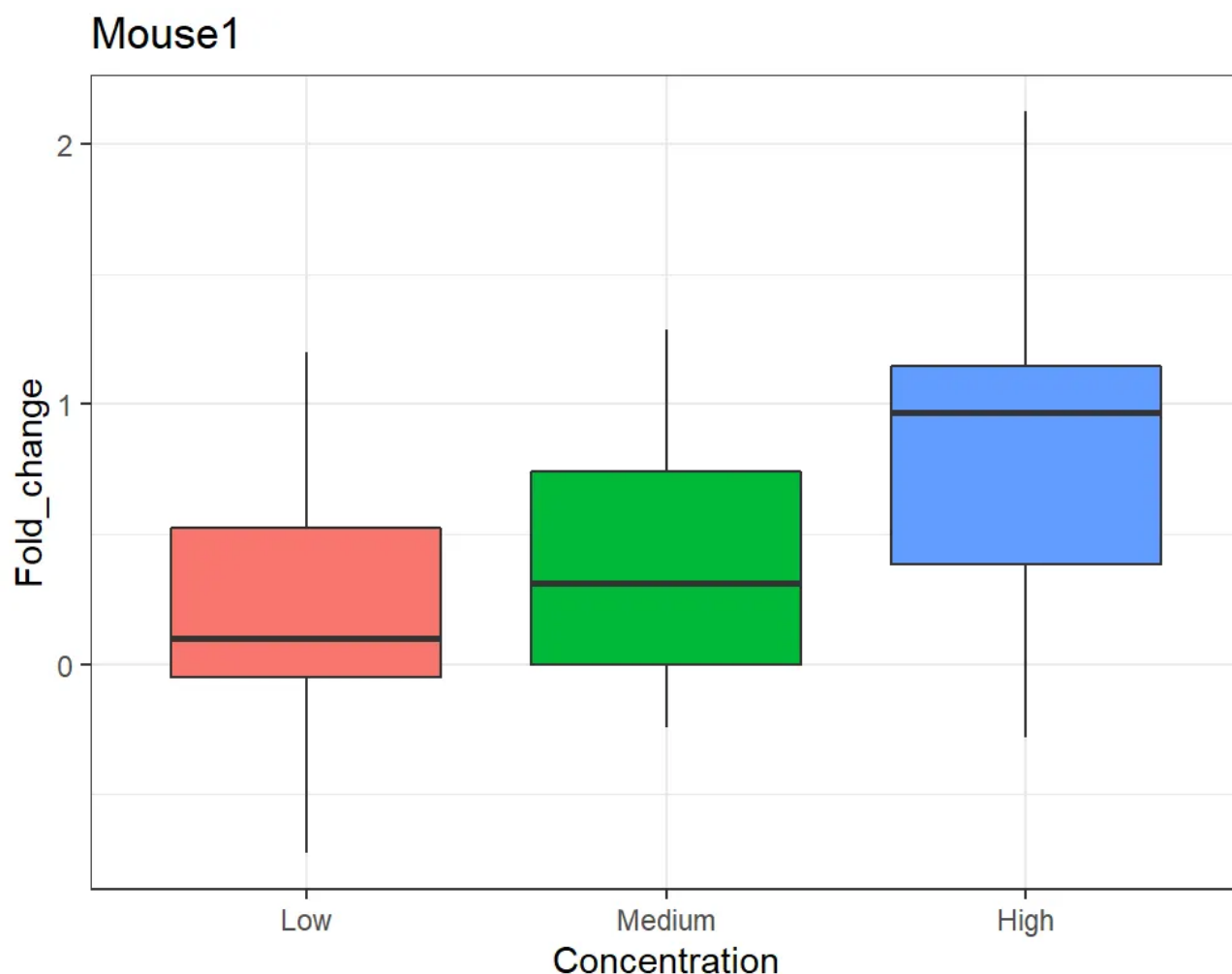
If we want to reduce the type I error, the TukeyHSD is recommend.

Compare with the baseline

If we want to compare the group with the baseline, for example if we set mouse1-low as the baseline and we want to compare the following combinations:

Mouse1-Low vs Mouse1-Medium

Mouse1-Low vs Mouse1-High



If I want to be very conservative, instead of comparing them with simple t-test, I would subset the dataset first and compare then with TukeyHSD.

```
subset_data = data %>% dplyr::filter(Mouse == "Mouse1")  
TukeyHSD(aov(Fold_change ~ Concentration, subset_data))
```



```
Tukey multiple comparisons of means
95% family-wise confidence level
```

```
Fit: aov(formula = Fold_change ~ Concentration, data = subset_data)
```

```
$Concentration
```

```
      diff      lwr      upr      p adj
Medium-Low 0.1402244 -0.3583714 0.6388201 0.7758651
High-Low    0.5660740  0.1554330 0.9767151 0.0046609
High-Medium 0.4258497 -0.1085783 0.9602776 0.1419619
```

Compare the pvalue between t-test and TukeyHSD

P-value	T-test	TukeyHSD
Medium-Low	0.4854	0.7758651
High-Low	0.003886	0.0046609

Take Home Message

1. Visualize your data before running the analysis
2. Find the best model to fit your data
3. Post-hoc analysis is to find out the difference between two groups
4. Using TukeyHSD instead of T-test to control the multiple correction

Resource

An introduction to the two-way ANOVA

ANOVA (Analysis of Variance) is a statistical test used to analyze the difference between the means of more than two...

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